

Mounica Paladugu

meismounica079@gmail.com | 206-591-5835 | Mountain View, CA | [linkedin.com/in/mounica-paladugu](https://www.linkedin.com/in/mounica-paladugu) | [mounica-000.github.io](https://github.com/mounica-000)

WORK EXPERIENCE

Software Engineer, Full Stack

Apr 2026 – Present

Saayam For All

Remote, CA

- Developing the distributed **backend REST API** microservices using Java, Spring Boot, Docker, and deploying them in Kafka, AWS EKS (Kubernetes) environments.
- Fixing existing issues on the webapp and working collaboratively on implementing features for the 1.0 version release.
- Writing unit tests and automated test cases for the developed features and fixed issues.

Software Engineer

Oct 2025 – Mar 2026

Develop For Good

Remote, CA

- Led **end-to-end** rebuild of a food recovery platform (React, TypeScript, Python, FastAPI), replacing Wix with a scalable application supporting donations, volunteer coordination, and food distribution.
- Designed and integrated Airtable as the database to enable **real-time updates** to the website and reduce content update time from days to seconds without developers involvement for the **nontechnical client**.
- Implemented a **Stripe-based** payment system, replacing manual QR-code donations and creating seamless payment flow to support operational costs.

Software Engineer Intern

Mar 2023 – Sep 2023

Rajant Corporation

Malvern, PA

- Architected a **scalable** backend for high-throughput telemetry metrics, migrating to **InfluxDB** to reduce query latency ~6x (350–500ms → 50–90ms) and improving system **observability** for large-scale data ingestion.
- Designed and deployed a **RESTful API (Python, Docker)** with **10+ endpoints** supporting multi-resolution health data aggregations, enabling seamless **system integration** with hospital records and an internal AI platform.
- Automated a real-time **data pipeline in Python** that watched a local directory and propagated changes to **MinIO S3 cloud storage** automatically, eliminating **5+ hours of manual work per week**.

Product Development Engineer Intern

Mar 2022– Sep 2022

Pression LLC

Coatesville, PA

- Automated the test data pipeline in **Python**, reducing file storage costs by 25% and cutting data loss by 50%.
- Led the medical device transition from an unstable v1 prototype to a **reliable manufacturing-ready v2**, integrating **embedded C firmware** with system UI software, achieved **zero field failures** across all test scenarios, and enabled the company to pursue **FDA clearance filing**.
- Refactored and debugged **2000+ lines of broken UI code** and authored comprehensive **system documentation** for 2x quicker onboarding of new interns.

SKILLS

Programming Languages: Python, Java, C, JavaScript/ TypeScript, SQL, Bash, HTML/CSS

Frameworks & Libraries: FastAPI, Node.js, React, Express, Spring Boot, JUnit

Databases & Tools: PostgreSQL, MySQL, InfluxDB, Git, Docker, Postman, AWS, Kubernetes

Systems & Practices: Linux/Unix, CI/CD, Agile, DevOps, SDLC, Jira, AI

PROJECTS

GitHub Issue Root Cause Agent | *Python, FastAPI, ChromaDB, Groq API, Streamlit UI* | [GitHub](#)

Mar 2026

- Built an **LLM-powered agent** with a **modular RAG pipeline**, pairing **ChromaDB vector search** with **Llama 3.3** to automate root cause analysis end-to-end.
- Achieved **89% retrieval precision** across paraphrased query variants by implementing **cosine similarity embeddings** to surface the top-3 most relevant past issues per bug report.
- Shipped a **REST API (FastAPI)** backed by **Llama 3.3 70b via Groq**, returning structured fix suggestions grounded in retrieved context, with a **Streamlit** demo UI.

Virtual Pet | *TypeScript, React, HTML/CSS, MySQL, Node.js*

Mar – May 2025

- Created a **full-stack real-time application** with **WebSocket-based** communication, **maintainable** normalized database schema, and **TypeScript** for end-to-end type safety.
- Deployed on a **Linux** (Linode) cloud server using **React, TypeScript, and MySQL**; wrote **unit and integration tests** to ensure code reliability.

EDUCATION

Drexel University

Philadelphia, PA

Bachelor of Science in Computer Science, Minor in Mathematics

Sep 2020 – Jun 2025